

Safety Data Sheet
acc. to OSHA HCS

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Printing date 09/16/2025
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Version number: 9.03

1 Identification

- **Product identifier**
- **Trade name:** Picric Acid moistened with ~ 33% of H₂O
- **Article number:** 1048
- **Application of the substance / the mixture** Laboratory chemicals
- **Details of the supplier of the safety data sheet**
- **Manufacturer/Supplier:**
PANREAC QUIMICA S.L.U.
C/Garraf 2
Polígono Pla de la Bruguera
E-08211 Castellar del Vallès (Barcelona)
Tel. (+34) 937 489 400
Fax. (+34) 937 489 401
e-mail: product.safety@itwreagents.com
- **Information department:** email: product.safety@panreac.com
- **Emergency telephone number:**
Einheitliche Notrufnummer: 112 (EU)
Tel.: (+34) 937 489 499

2 Hazard(s) identification

- **Classification of the substance or mixture**
Flammable Solids 2 H228 Flammable solid.
Acute Toxicity - Oral 3 H301 Toxic if swallowed.
Acute Toxicity - Dermal 3 H311 Toxic in contact with skin.
Acute Toxicity - Inhalation 3 H331 Toxic if inhaled.

- **Label elements**
- **GHS label elements**
The product is classified and labeled according to the Globally Harmonized System (GHS).
- **Hazard pictograms**



GHS02 GHS06

- **Signal word** Danger
- **Hazard-determining components of labeling:**
2,4,6-trinitrophenol

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US

Trade name: Picric Acid moistened with ~ 33% of H2O

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· **Hazard statements**

H228 Flammable solid.
H301+H311+H331 Toxic if swallowed, in contact with skin or if inhaled.

· **Precautionary statements**

P210 Keep away from heat/sparks/open flames/hot surfaces. - No smoking.
P240 Ground/bond container and receiving equipment.
P241 Use explosion-proof electrical/ventilating/lighting/equipment.
P261 Avoid breathing dust/fume/gas/mist/vapors/spray
P264 Wash thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P271 Use only outdoors or in a well-ventilated area.
P280 Wear protective gloves/protective clothing/eye protection/face protection.
P301+P310 If swallowed: Immediately call a poison center/doctor.
P321 Specific treatment (see on this label).
P330 Rinse mouth.
P302+P352 If on skin: Wash with plenty of water.
P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P311 Call a poison center/doctor.
P312 Call a poison center/doctor if you feel unwell.
P361+P364 Take off immediately all contaminated clothing and wash it before reuse.
P370+P378 In case of fire: Use CO2, powder or water spray to extinguish.
P403+P233 Store in a well-ventilated place. Keep container tightly closed.
P405 Store locked up.
P501 Dispose of contents/container in accordance with local/regional/national/international regulations.

· **Classification system:**

· **NFPA ratings (scale 0 - 4)**



Health = 3
Fire = 1
Reactivity = 0

· **HMIS-ratings (scale 0 - 4)**



Health = 2
Fire = 1
Reactivity = 0

· **Other hazards**

· **Results of PBT and vPvB assessment**

- **PBT:** Not applicable.
- **vPvB:** Not applicable.

3 Composition/information on ingredients

· **Chemical characterization: Mixtures**

· **Description:** Mixture of the substances listed below with nonhazardous additions.

· **Dangerous components:**

88-89-1	2,4,6-trinitrophenol	>50-<100%
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US

(Contd. on page 3)

Trade name: Picric Acid moistened with ~ 33% of H2O

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4 First-aid measures

- **Description of first aid measures**

- **General information:**

Immediately remove any clothing soiled by the product.
Remove breathing apparatus only after contaminated clothing have been completely removed.
In case of irregular breathing or respiratory arrest provide artificial respiration.
Involve doctor immediately.

- **After inhalation:**

Supply fresh air or oxygen; call for doctor.
In case of unconsciousness place patient stably in side position for transportation.
If not breathing, if breathing is irregular or if respiratory arrest occurs, provide artificial respiration or oxygen by trained personnel.

- **After skin contact:**

Call a doctor immediately.
Immediately wash with water and soap and rinse thoroughly.

- **After eye contact:** Rinse opened eye for several minutes under running water. Then consult a doctor.

- **After swallowing:**

make victim drink water (maximum of 2 drinking glasses)
Laxative: Sodium sulfate (1 tablespoon/ 1/4 L water)
activated charcoal (20 - 40 g in 10 % slurry)
Inform doctor. Do not give milk or fatty oils.
Do not induce vomiting; immediately call for medical help.

- **Information for doctor:**

- **Most important symptoms and effects, both acute and delayed**

No further relevant information available.

- **Indication of any immediate medical attention and special treatment needed**

No further relevant information available.

5 Fire-fighting measures

- **Extinguishing media**

- **Suitable extinguishing agents:**

Water
Alcohol resistant foam
Use fire fighting measures that suit the environment.

- **For safety reasons unsuitable extinguishing agents:**

Water with full jet
None

- **Special hazards arising from the substance or mixture**

In case of fire, the following can be released:
Nitrogen oxides (NOx)
Carbon monoxide and carbon dioxide
Combustible.
Possible self-ignition without protective liquid.
Risk of explosion in the event of shock or friction.

- **Advice for firefighters**

- **Protective equipment:**

Mouth respiratory protective device.
Wear self-contained respiratory protective device.
Wear fully protective suit.

- **Additional information**

Cool endangered receptacles with water spray.

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Trade name: Picric Acid moistened with ~ 33% of H2O

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Dispose of fire debris and contaminated fire fighting water in accordance with official regulations.

6 Accidental release measures

- **Personal precautions, protective equipment and emergency procedures**
Avoid formation of dust.
Wear protective equipment. Keep unprotected persons away.
Avoid substance contact.
Ensure adequate ventilation.
- **Environmental precautions:** Do not allow to enter sewers/ surface or ground water.
- **Methods and material for containment and cleaning up:**
Pick up mechanically.
Avoid formation of dust.
Do not allow to dry out
Collect up carefully, striving to keep dampened.
Dispose contaminated material as waste according to section 13.
Ensure adequate ventilation.
Clean up affected area.
- **Reference to other sections**
See Section 7 for information on safe handling.
See Section 8 for information on personal protection equipment.
See Section 13 for disposal information.
- **Protective Action Criteria for Chemicals**

· PAC-1:

88-89-1	2,4,6-trinitrophenol	0.3 mg/m ³
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· PAC-2:

88-89-1	2,4,6-trinitrophenol	17 mg/m ³
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· PAC-3:

88-89-1	2,4,6-trinitrophenol	100 mg/m ³
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7 Handling and storage

- **Handling:**
- **Precautions for safe handling**
Thorough dedusting.
Ensure good ventilation/exhaustion at the workplace.
Open and handle receptacle with care.
Any deposit of dust which cannot be avoided must be regularly removed.
- **Information about protection against explosions and fires:**
Keep ignition sources away - Do not smoke.
Protect from heat.
Protect against electrostatic charges.
Keep respiratory protective device available.
- **Conditions for safe storage, including any incompatibilities**
- **Storage:**
- **Requirements to be met by storerooms and receptacles:** Store in a cool location.
- **Information about storage in one common storage facility:** Not required.
- **Further information about storage conditions:**
Keep receptacle tightly sealed.
Store in cool, dry conditions in well sealed receptacles.
Protect from heat and direct sunlight.

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(Contd. of page 4)

- Store receptacle in a well ventilated area.
- Store under lock and key and with access restricted to technical experts or their assistants only.
- **Recommended storage temperature:** +2°C - +25°C
- **Storage class:** 4.1 B
- **Specific end use(s)** No further relevant information available.

8 Exposure controls/personal protection

- **Additional information about design of technical systems:** No further data; see section 7.
- **Control parameters**

• **Components with limit values that require monitoring at the workplace:**

88-89-1 2,4,6-trinitrophenol

PEL	Long-term value: 0.1 mg/m ³ Skin
REL	Short-term value: 0.3 mg/m ³ Long-term value: 0.1 mg/m ³ Skin
TLV	Long-term value: 0.1 mg/m ³

- **Additional information:** The lists that were valid during the creation were used as basis.
- **Exposure controls**
- **Personal protective equipment:**
- **General protective and hygienic measures:**
 - Keep away from foodstuffs, beverages and feed.
 - Immediately remove all soiled and contaminated clothing.
 - Wash hands before breaks and at the end of work.
 - Store protective clothing separately.
 - Vacuum clean contaminated clothing. Do not blow or brush off contamination.
 - Avoid contact with the eyes and skin.
- **Breathing equipment:**
 - Filter P3
 - In case of brief exposure or low pollution use respiratory filter device. In case of intensive or longer exposure use respiratory protective device that is independent of circulating air.
 - Required when dusts are generated.
- **Protection of hands:**



Protective gloves

The glove material has to be impermeable and resistant to the product/ the substance/ the preparation. Selection of the glove material on consideration of the penetration times, rates of diffusion and the degradation

- **Material of gloves**

The selection of the suitable gloves does not only depend on the material, but also on further marks of quality and varies from manufacturer to manufacturer. As the product is a preparation of several substances, the resistance of the glove material can not be calculated in advance and has therefore to be checked prior to the application.
- **Penetration time of glove material**

The exact break through time has to be found out by the manufacturer of the protective gloves and has to be observed.
- **For the permanent contact gloves made of the following materials are suitable:**
 - Nitrile rubber, NBR

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Recommended thickness of the material: ≥ 0.11 mm

Value for the permeation: Level ≥ 480 min

· **As protection from splashes gloves made of the following materials are suitable:**

Nitrile rubber, NBR

Recommended thickness of the material: ≥ 0.11 mm

Value for the permeation: Level ≥ 480 min

· **Eye protection:**



Tightly sealed goggles

· **Body protection:**

Use protective suit.

Full head, face and neck protection

9 Physical and chemical properties

· **Information on basic physical and chemical properties**

· **General Information**

· **Appearance:**

Form:

Solid

Color:

Yellow

· **Odor:**

Odorless

· **Odor threshold:**

Not determined.

· **pH-value:**

Not applicable.

· **Change in condition**

Melting point/Melting range:

122.5 °C (252.5 °F)

Boiling point/Boiling range:

>300 °C (>572 °F)

· **Flash point:**

150 °C (302 °F)

· **Flammability:**

Not determined.

Not applicable.

Highly flammable.

· **Decomposition temperature:**

Not determined.

· **Ignition temperature:**

Product is not selfigniting.

· **Danger of explosion:**

Not determined.

· **Explosion limits:**

Lower:

Not determined.

Upper:

Not determined.

· **Vapor pressure at 20 °C (68 °F):**

23 hPa (17.3 mm Hg)

· **Density at 20 °C (68 °F):**

1.8 g/cm³ (15.021 lbs/gal)

· **Relative density**

Not determined.

· **Vapor density**

Not applicable.

· **Evaporation rate**

Not applicable.

· **Solubility in / Miscibility with**

Water:

Insoluble.

· **Partition coefficient (n-octanol/water):** Not determined.

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· Viscosity:	
Dynamic:	Not applicable.
Kinematic:	Not applicable.
· Solvent content:	
Water:	33.0 %
VOC content:	0.00 %
Solids content:	100.0 %
· Other information	No further relevant information available.

10 Stability and reactivity

- **Reactivity** Danger of explosion when dry.
- **Chemical stability**
- **Thermal decomposition / conditions to be avoided:**
No decomposition if used according to specifications.
- **Possibility of hazardous reactions** No dangerous reactions known.
- **Conditions to avoid**
Heating
Danger of explosion when dry.
- **Incompatible materials:**
alkaline earth metals
alkaline earth compounds
alkali metals
alkali salts
fluorine
- **Hazardous decomposition products:** In the event of fire: See chapter 5

11 Toxicological information

- **Information on toxicological effects**
- **Acute toxicity:**
- **LD/LC50 values that are relevant for classification:**
Quantitative data on the toxicological effect of this product are not available.

Components	Type	Value	Species
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ATE (Acute Toxicity Estimate)

Oral	LD50	149 mg/kg
Dermal	LD50	448 mg/kg
Inhalative	LC50/4 h	0.75 mg/l

88-89-1 2,4,6-trinitrophenol

Oral	LD50	100 mg/kg (ATE)
Dermal	LD50	300 mg/kg (ATE)
Inhalative	LC50/4 h	0.5 mg/l (ATE)

- **Primary irritant effect:**
- **on the skin:** No irritant effect.
- **on the eye:** No irritating effect.
- **Sensitization:** No sensitizing effects known.

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· **Additional toxicological information:**

The product shows the following dangers according to internally approved calculation methods for preparations:
Toxic

· **Carcinogenic categories**

· **IARC (International Agency for Research on Cancer)**

None of the ingredients is listed.

· **NTP (National Toxicology Program)**

None of the ingredients is listed.

· **OSHA-Ca (Occupational Safety & Health Administration)**

None of the ingredients is listed.

12 Ecological information

· **Toxicity**

· **Aquatic toxicity:** No further relevant information available.

· **Persistence and degradability** No further relevant information available.

· **Behavior in environmental systems:**

· **Bioaccumulative potential** No further relevant information available.

· **Mobility in soil** No further relevant information available.

· **Additional ecological information:**

· **General notes:**

Do not allow product to reach ground water, water course or sewage system.

Water hazard class 2 (Self-assessment): hazardous for water

Danger to drinking water if even small quantities leak into the ground.

· **Results of PBT and vPvB assessment**

· **PBT:** Not applicable.

· **vPvB:** Not applicable.

· **Other adverse effects** No further relevant information available.

13 Disposal considerations

· **Waste treatment methods**

· **Recommendation:**

Chemicals must be disposed of in compliance with the respective national regulations.

Must not be disposed of together with household garbage. Do not allow product to reach sewage system.

· **Uncleaned packagings:**

· **Recommendation:**

Disposal must be made according to official regulations.

Packagings that cannot be cleansed are to be disposed of in the same manner as the product.

14 Transport information

· **UN-Number**

· **DOT, ADR, IMDG, IATA**

UN1344

· **UN proper shipping name**

· **DOT**

Trinitrophenol, wetted

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· ADR, IMDG, IATA		TRINITROPHENOL (PICRIC ACID), WETTED
· Transport hazard class(es)		
· DOT		
		
· Class	4.1 Flammable solids, self-reactive substances and solid desensitised explosives	
· Label	4.1	
· ADR		
		
· Class	4.1 (D) Flammable solids, self-reactive substances and solid desensitised explosives	
· Label	4.1	
· IMDG, IATA		
		
· Class	4.1 Flammable solids, self-reactive substances and solid desensitised explosives	
· Label	4.1	
· Packing group		
· DOT, ADR, IMDG, IATA	I	
· Environmental hazards:	Not applicable.	
· Special precautions for user	Warning: Flammable solids, self-reactive substances and solid desensitised explosives	
· Hazard identification number (Kemler code):	-	
· EMS Number:	F-B,S-J	
· Stowage Category	B	
· Stowage Code	SW2 Clear of living quarters.	
· Transport in bulk according to Annex II of MARPOL73/78 and the IBC Code	Not applicable.	
· Transport/Additional information:		
· ADR		
· Excepted quantities (EQ)	Code: E0 Not permitted as Excepted Quantity	
· IMDG		
· Limited quantities (LQ)	0	
· Excepted quantities (EQ)	Code: E0 Not permitted as Excepted Quantity	

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Trade name: Picric Acid moistened with ~ 33% of H2O

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· **UN "Model Regulation":** UN 1344 TRINITROPHENOL (PICRIC ACID), WETTED, 4.1, I

15 Regulatory information

· **Safety, health and environmental regulations/legislation specific for the substance or mixture**
No further relevant information available.

· **Sara**

· **Section 355 (extremely hazardous substances):**

None of the ingredients is listed.

· **Section 313 (Specific toxic chemical listings):**

88-89-1 2,4,6-trinitrophenol

· **TSCA (Toxic Substances Control Act):**

All components have the value ACTIVE.

· **Hazardous Air Pollutants**

None of the ingredients is listed.

· **Proposition 65**

· **Chemicals known to cause cancer:**

None of the ingredients is listed.

· **Chemicals known to cause reproductive toxicity for females:**

None of the ingredients is listed.

· **Chemicals known to cause reproductive toxicity for males:**

None of the ingredients is listed.

· **Chemicals known to cause developmental toxicity:**

None of the ingredients is listed.

· **Carcinogenicity categories**

· **EPA (Environmental Protection Agency)**

None of the ingredients is listed.

· **TLV (Threshold Limit Value)**

None of the ingredients is listed.

· **NIOSH-Ca (National Institute for Occupational Safety and Health)**

None of the ingredients is listed.

· **GHS label elements**

The product is classified and labeled according to the Globally Harmonized System (GHS).

· **Hazard pictograms**



GHS02 GHS06

· **Signal word** Danger

· **Hazard-determining components of labeling:**

2,4,6-trinitrophenol

(Contd. on page 11)

US

Trade name: Picric Acid moistened with ~ 33% of H2O

(Contd. of page 10)

· **Hazard statements**

H228 Flammable solid.
H301+H311+H331 Toxic if swallowed, in contact with skin or if inhaled.

· **Precautionary statements**

P210 Keep away from heat/sparks/open flames/hot surfaces. - No smoking.
P240 Ground/bond container and receiving equipment.
P241 Use explosion-proof electrical/ventilating/lighting/equipment.
P261 Avoid breathing dust/fume/gas/mist/vapors/spray
P264 Wash thoroughly after handling.
P270 Do not eat, drink or smoke when using this product.
P271 Use only outdoors or in a well-ventilated area.
P280 Wear protective gloves/protective clothing/eye protection/face protection.
P301+P310 If swallowed: Immediately call a poison center/doctor.
P321 Specific treatment (see on this label).
P330 Rinse mouth.
P302+P352 If on skin: Wash with plenty of water.
P304+P340 IF INHALED: Remove person to fresh air and keep comfortable for breathing.
P311 Call a poison center/doctor.
P312 Call a poison center/doctor if you feel unwell.
P361+P364 Take off immediately all contaminated clothing and wash it before reuse.
P370+P378 In case of fire: Use CO2, powder or water spray to extinguish.
P403+P233 Store in a well-ventilated place. Keep container tightly closed.
P405 Store locked up.
P501 Dispose of contents/container in accordance with local/regional/national/international regulations.

· **Chemical safety assessment:** A Chemical Safety Assessment has not been carried out.

16 Other information

This information is based on our present knowledge. However, this shall not constitute a guarantee for any specific product features and shall not establish a legally valid contractual relationship.

· **Contact:**

· **Date of preparation / last revision** 09/16/2025 / 9.02

· **Abbreviations and acronyms:**

ADR: Accord relatif au transport international des marchandises dangereuses par route (European Agreement Concerning the International Carriage of Dangerous Goods by Road)
IMDG: International Maritime Code for Dangerous Goods
DOT: US Department of Transportation
IATA: International Air Transport Association
EINECS: European Inventory of Existing Commercial Chemical Substances
ELINCS: European List of Notified Chemical Substances
CAS: Chemical Abstracts Service (division of the American Chemical Society)
NFPA: National Fire Protection Association (USA)
HMIS: Hazardous Materials Identification System (USA)
VOC: Volatile Organic Compounds (USA, EU)
LC50: Lethal concentration, 50 percent
LD50: Lethal dose, 50 percent
PBT: Persistent, Bioaccumulative and Toxic
vPvB: very Persistent and very Bioaccumulative
NIOSH: National Institute for Occupational Safety
OSHA: Occupational Safety & Health
TLV: Threshold Limit Value
PEL: Permissible Exposure Limit
REL: Recommended Exposure Limit
Flammable Solids 2: Flammable solids – Category 2
Acute Toxicity - Oral 3: Acute toxicity – Category 3

· *** Data compared to the previous version altered.**